|INCH-POUND|

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13 January 1992
SUPERSEDING
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5 March 1985

MILITARY SPECIFICATION

PUNCHING MACHINE, METAL, POWER DRIVEN, SINGLE END TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers power driven metal punching machines of the single end type with equipment and accessories specified herein.

1.2 Classification. The machine shall be of the following sizes. The size to be furnished shall be as specified (see 3.5 and 6.2).

Size	Punch Throat Depth (Inches)	Nominal Punch Capacity (Tons)
11	30	30
12	30	40
13	30	50

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis, TN 38114-5213, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.2 - Hydraulic Presses - Safety Requirements for the Construction, Care, and Use. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

NATIONAL FLUID POWER ASSOCIATION, INC. (NFPA)

NFPA/JIC T2.24.1 - Hydraulic fluid power - Systems standard for industrial machinery.

(Application for copies should be addressed to the National Fluid Power Association Inc., 3333 N. Mayfair Road, Milwaukee, WI 53222-3291.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall accomplish punching operations using a powered single end punch. The frame of the machine shall be an integral unit designed to rest on a flat mounting surface. Provisions suitable for hoisting the machine shall be provided. The machine shall

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include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and electrical controls shall be located convenient to the operator's work station. Controls shall include means for starting and stopping the machine and other necessary controlling devices. All operations of the machine except punch spotting shall be controlled by an anti-repeat foot switch.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform with all requirements of OSHA 29 CFR 1910 and ANSI B11.2. If a conflict arises between OSHA 29 CFR 1910 and ANSI B11.2, OSHA 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2).

3.2.5.1 Covers, guards and other safety devices. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the machine. All guards shall allow access to the guarded parts.

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3.2.5.2 Noise level. Unless otherwise specified (see 6.2), noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Environmental protection. During the manufacture, service, transportation, or storage of the machine, and its components, the machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions in effect on the date of the contract. The operation of the machine at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.8 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, filler caps, and the reservoir drain plug shall be accessible.

3.2.9 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein (see 6.2), the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

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3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a frame, table, machine drive, punch holder, die holder, stripper, tool storage cabinet, front to back and side gauge, and electrical requirements.

3.4.1 Frame. The frame structure shall be either cast iron, cast steel, or a fabricated steel weldment of the mass, weight, and rigidity to withstand stresses produced when operating the machine at its maximum capacity without permanent deformation or failure. The frame shall be of "C" style configuration, stress relieved, with the overhang and base forming a workpiece opening or throat. The depth of the throat shall be as specified for the machine sizes listed in table I. The frame enclosure shall have covered access openings of the size required to facilitate servicing, maintenance, and adjustments of all components contained therein. The frame shall have provisions for securing the machine to a mounting surface. An eye or attachment point shall be provided at or near its balance point on the frame for lifting the machine. The eye or lift point shall be capable of supporting five times the weight of the machine, and its location shall not cause distortion to the frame when the machine is suspended.

3.4.2 Table. The machine shall be provided with a table for supporting the workpiece. The table top shall be of ductile iron, plate construction or fabricated steel (stress relieved) construction, and of sufficient size and strength to support the workpiece. The positioning area of the table shall accommodate the workpiece in the punching operation. Unless otherwise specified (see 6.2), the workpiece positioning area on the table shall conform to the requirements of table I.

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TABLE I. Sizes and capacities.

Characteristic	Size 11	Size 12	Size 13
Rated capacity (tons)	30	40	50
Throat depth (inches)	30	30	30
Stroke length (inches)	1/2	1/2	1/2
Strokes per minute	150	110	80
Hole punching capacity in 7 ga. (.1793) inch thick mild steel diameter (inches)	1-1/4	2-13/16	3-1/2
Maximum hole diameter capacity of machine (inches)	3-1/2	5	5
Workpiece positioning area (inches)			
side to side	30	30	72
front to back	30	30	30

NOTE: Sizes and capacities are based on using 50,000 PSI shear strength steel, 20 percent minimum die clearance, and flat punches. Requirements shall be not less than those stated for the size machine ordered.

3.4.3 Machine drive. Unless otherwise specified, the punching action of the machine shall be accomplished by either a mechanical or hydraulic type drive. When only one type drive is acceptable, the particular drive type shall be as specified (see 6.2). The drive shall be designed to assure that sufficient energy is developed for accomplishing the performance requirements specified herein. Unless otherwise specified (see 6.2), the drive shall be controlled by a foot pedal type control.

3.4.3.1 Mechanical type drive. The mechanical type drive when furnished shall be provided with an electric motor driven flywheel for transmitting the required punching force to a machine ram through a drive shaft. The drive shall include clutch and brake mechanisms for controlling the punching action. The clutch mechanism shall be arranged to provide both single and continuous stroke operation.

3.4.3.2 Hydraulic type drive. The hydraulic type drive when furnished, shall conform to NFPA/JIC T2.24.1. The hydraulic drive shall include a hydraulic pump, fluid reservoir, ram cylinder unit, and connecting hoses or tubing. The hydraulic drive shall be kept clean by means of filters which ensure proper operation of the system. The hydraulic drive shall operate on a non-flammable fluid. The hydraulic type drive shall be designed to provide both single and continuous stroke operation.

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3.4.3.2.1 Hydraulic pump. The hyudraulic fluid pump shall be of the motor driven type. The pump motor shall conform to the requirements of 3.4.11. The pump shall be provided with hydraulic fluid control devices which shall include a directional control valve and a pressure relief valve. The pressure relief valve shall be adjusted with a relief setting of not more than 10 percent higher than the highest working pressure of the system.

3.4.3.2.2 Reservoir. The hydraulic fluid reservoir shall have sufficient volumetric capacity to supply the pump volume and circuit operating requirements, and maintain the fluid temperature within its recommended temperature range, while operating the machine at maximum capacity. The reservoir shall be provided with an accessible pump suction filter, filter strainer, sight level gauge, magnetic drain plug and a covered access cleanout opening.

3.4.3.2.3 Ram cylinder unit. The ram cylinder unit shall consist of a double acting cylinder connected to the frame with a piston connected to the ram. The cylinder shall be sealed against leakage of hydraulic fluid. The ram cylinder unit shall provide a means for incrementally lowering the punch for positioning over the work.

3.4.3.2.4 Connecting hose or tubing. Hydraulic fluid transfer between the pump and the ram cylinders shall be accomplished through connecting hose or tubing. The hose or tubing shall conform to the requirements of NFPA/JIC T2.24.1.

3.4.4 Punch holder. The machine shall be provided with a punch holder for holding the punch in punching position under the ram cylinder unit. The punch holders shall be constructed of alloy steel and securely hold the punch in position. The punch holders of shaped type punches shall be keyed to assure positive alignment with the punches and dies. The punch holders shall be designed to accommodate standard round and shaped hole punches.

3.4.5 Die holder. The machine shall be provided with a die holder for holding the die in position. The die holders shall be constructed of hardened steel and precision ground to securely hold the dies in such a manner to prevent the dies from turning. The die holders shall be designed to accommodate standard round and shaped hole dies.

3.4.6 Stripper. The machine shall be provided with a stripper or other means for automatically releasing the workpiece from the punch when the punch is withdrawn from the die.

3.4.7 Punches and dies. When required, punches and dies of sizes and shapes specified (see 6.2) shall be furnished. The punches and dies shall be tool steel, hardened, and ground for accuracy. Each punch and die shall be permanently marked for size and identification.

3.4.8 Tool storage cabinet. The machine shall be provided with a tool storage cabinet, located either on the machine or furnished as a separate unit. The cabinet shall be constructed to enclose and accommodate at least 8 cubic feet of tooling. The cabinet shall contain shelving and/or tool holders as specified by the procuring activity (see 6.2).

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3.4.9 Front to back and side gauges. Unless otherwise specified (see 6.2), the machine shall be furnished with front to back gauge and a side gauge for aligning the workpiece under the ram and punch. The gauge shall be provided with a guide bar for controlling workpiece alignment across the table from front to back along the Y axis. The guide bar shall move across the table in machine ways or by other means to maintain a constant plane, parallel with the X axis throughout the adjustment range. The guide bar shall be provided with not less than 9 workpiece stops for aligning the workpiece from the center of the ram and punch to either side along the X axis. The workpiece stops shall be arranged to slide along and be locked to the guide bar at intervals for programming required hole locations. The guide bar and workpiece stops shall be provided with micrometer dials or micrometer gauges for measuring workpiece settings of 0.001 inch or less in both the X and Y axes. Each axis of the gauge shall be provided with scales, permanently marked in graduated increments, for measuring the distance from the center of the ram and punch through the adjustment range for the X and Y axes as specified in table I. A device shall be provided for locking the gauge in position at any point within the adjustment ranges.

3.4.10 Stylus type workpiece position gauge. When specified (see 6.2), the machine shall be furnished with a stylus type workpiece positioning gauge for aligning the workpiece under the ram and punch. The gauge shall be a duplicator style device with a template table, workpiece holding carriage, and hand controlled stylus for locating the punching points on the templates and actuating the machine drive for punching. The table shall have provisions for securely holding the template. The carriage shall be provided with not less than two clamps for holding the workpiece for punching operations.

3.4.11 Electrical requirements. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz (Hz), dual 230/460-volts. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired as specified (see 6.2).

3.4.11.1 Motors. Motors shall meet the requirements of 3.4.11. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.11.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built in lighting system is provided, the adjustable light described above shall not be required.

3.4.12 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of

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not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I for the size machine specified. Machine size and capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2.), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The machine shall be capable of performing punching operations in accordance with the requirements of table I. The operations shall be performed with standard style tooling with no shear on the punches or dies. When performing the operations, there shall be no permanent deformation of the workpiece.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional and accessory equipment. Optional and accessory equipment shall be furnished as specified and fully described (see 6.2). The equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on the machine including all instruments, control panels, control units, charts, plates, and related attachments shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate or chart. When specified (see 6.2) a lubrication plate or chart shall be attached to each machine. If a chart is furnished it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets. Unless otherwise specified (see 6.2) the information provided on the chart or plate shall be as listed below.

Points of lubricant application
Type of lubricant
Servicing interval
SAE number or military or Federal specification for each lubricant (if applicable)

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3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.10.3 Hydraulic fluid requirements plate. When specified (see 6.2), a hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.11 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein.

3.12 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. Each machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.8.3 Performance tests. The machine shall be tested for conformance with 3.6 by punching capacity size holes in 7 gauge (.1793) inch thick mild steel. The machine shall punch not less than 12 holes each of the diameters specified in the material thickness and type as specified in table I for the size machine being tested. The holes shall be punched without deforming the test pieces. All performance tests shall be completed without making any adjustments to or performing any maintenance to the machine.

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4.8.3.1 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.4 Noise level test. Noise level measurements shall be made to ensure conformance with 3.2.5.2. Unless otherwise specified (see 6.2), the measurements shall be made at the operators work station and each corner of the machine. Noise emitted by the machine when operating under a no load condition shall not exceed 84 db when measured on the A scale of a type 1 or type 2 sound level meter at slow response. The test shall be conducted with the machine operating at intermediate and maximum speeds.

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

4.10 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.7. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are intended to incorporate simple design, easy set up, and rapid adjustment for use in short run punching operations. A variety of operations may be performed on the same machine without expensive tooling and movement of job to other machines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound) (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5.1).

h. Acceptable noise level, if diff_rent (see 3.2.5.2).

i. Environmental protection requirements exceeding EPA/OSHA requirements (see 3.2.7).

j. Specify surface finish requirements, if different (see 3.3.3).

k. Painting, if different (see 3.3.5.).

l. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

m. Table top positioning area, if different (see 3.4.2).

n. Specify drive type, if applicable (see 3.4.3).

o. Type of control, if different than foot pedal (see 3.4.3).

p. Punch and die sizes and shapes, when required (see 3.4.7).

q. Specify shelving or tool holders required (see 3.4.8).

r. Gauges, if different (see 3.4.9).

s. Stylus type workpiece positioning gauge, when specified (see 3.4.10).

t. Voltage requirements, if different (see 3.4.11).

u. Initial voltage, as specified (see 3.4.11).

v. Motors, if different (see 3.4.11.1).

w. Worklight, if required (see 3.4.11.2).

x. Hour meter, if required (see 3.4.12).

y. Size and capacity, if different (see 3.5).

z. Specify physical size limitation, if required (see 3.5.1).

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aa. Additional performance features, if required (see 3.6.1).

bb. Specify and fully describe optional equipment, if required (see 3.8).

cc. Specify applicable repair parts, if required (see 3.9).

dd. Lubrication plate, if required (see 3.10.1).

ee. Lubrication plate information, if different (see 3.10.1).

ff. Nameplate, if different (see 3.10.2).

gg. Hydraulic fluid requirements plate, when required (see 3.10.3).

hh. Hydraulic fluid requirements plate information, if different (see 3.10.3).

ii. Inspection conditions, if different (see 4.3).

jj. First article inspection, if different (see 4.4).

kk. Quality conformance inspection, if different (4.5).

ll. Acceptance test, if different (see 4.6).

mm. Additional performance tests, if required (see 4.8.3.1).

nn. Location of noise level test, if different (see 4.8.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language
N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification. Classification changes from the previous issue of this specification are as follows:

MIL-P-80072B	MIL-P-80072C
Size 1	Size 1 - deleted
2	2 - deleted
3	3 - deleted
4	4 - deleted
5	5 - deleted
6	6 - deleted
7	7 - deleted
8	8 - deleted
9	9 - deleted
10	10 - deleted
	11 - new
	12 - new
	13 - new

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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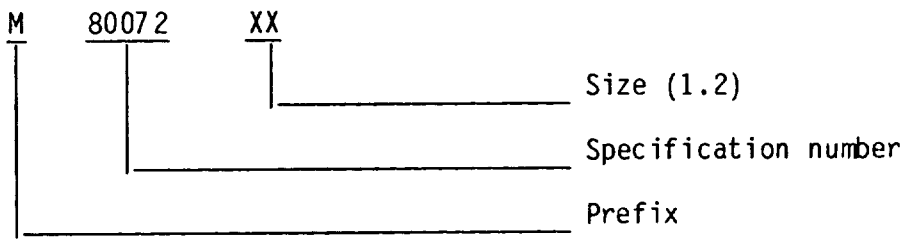
6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Training. Training required by the procuring activity should be provided as specified in the contract.

6.11 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.12 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.13 Part or Identifying Number (PIN). The PINs to be used for the machine acquired to this specification are created as follows:



6.14 Subject term Key word listing.

Punch
Punching Machine
Single End Punching Machine

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Air Force - 84
DLA - GS

User activities:
Navy - YD, MC

Preparing Activity:
DLA - IP

(Project 3445-0122)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-80072C	2. DOCUMENT DATE (YYMMDD) 92 01 13

3. DOCUMENT TITLE
PUNCHING MACHINE, METAL, POWER DRIVEN, SINGLE END TYPE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	(1) Commercial (901)-775-6269	(2) AUTOVON 683-6269
c. ADDRESS *(Include Zip Code)* 2163 Airways Blvd. Memphis, TN 38114-5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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